Goldman Sachs | Commodities Research

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Green Metals

Battery Metals Watch: No bottom in sight

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Structurally bearish fundamentals on oversupply. Battery metals prices have fallen significantly YTD, with China lithium carbonate prices declining 41%, nickel falling 29% and cobalt metal down 28%. Whilst this correction in prices has been predominantly tied to a sluggish start to the year in China's EV sector, we think that part of the sell-off also reflects respective markets' focus on strong supply trends across the complex. The already apparent fundamental shift from scarcity to a glut in the cobalt and nickel markets has led to continued price falls in search of tightening adjustments, whilst a sequential improvement in Chinese demand has led to a bounce in lithium prices where the oversupply is less pronounced for now. Going forward, on the demand side, we remain optimistic on EV sales in China for FY 2023 (8.5Mn units, +2Mn units y/y) on realised strategic EV price cuts, extension of tax exemptions and targeted stimulus support in rural areas. However, we expect supply to sequentially soften the lithium market into surplus over H2 and continue to oversupply both the cobalt and nickel markets. Similar to the latest nickel update, we increase surplus for cobalt starting 2023, whereas for lithium 2023 we modestly lower our surplus before moving into a high surplus phase over 2024-25. As a result, we expect battery metals prices to fall in H2, especially so for lithium where prices are still substantially above the top-end of cost curve which stands in contrast to an incremental necessity for supply rationing into 2024. We will, however, note that the long-term drivers of demand growth remain resilient as we raise our demand expectation from energy storage systems on renewable installation acceleration, particularly in China.

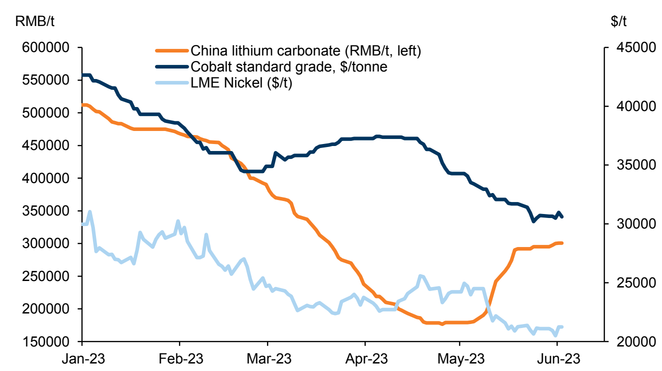
Further medium-term downside stays for lithium. Over the past month, lithium prices have bounced back on signs of recovery in the Chinese EV market. Whilst we expect recovering demand to support lithium prices in the near-term, we continue to see a supply-led correction over H2. On our estimates, global spodumene is still the main driver of incremental raw material supply growth, on track to add 500kt LCE cumulative capacity over 2023-24, and China's lepidolite capacity will likely rise over 200kty LCE by end of next year, with total refined supply growing 321kty LCE on average. This compares against an average 226kty LCE of battery demand growth over the same period. In this context, we continue to see substantial downside to lithium prices from current spot levels, with our 12m target for China battery grade lithium carbonate (ex-VAT) at $15,000/t LCE but raise the 2024 average to $15,331/t LCE (vs. $11,000/t LCE prev.) on the higher cost for non-integrated projects and the lengthy quality check timeline for new feedstock by converters. However, if the boost driven by government measures and lower EV prices do not generate high EV sales in China over the coming months, it will lead to an earlier sell-off in lithium prices exacerbated by high lithium inventories in EV supply chain, in our view. Equally, we note that the refining shift from carbonate to hydroxide will likely require an attractive conversion premium and introduce a 12m target for Asia CIF lithium hydroxide at $19,000/t. Whilst we think prices will move to a floor pricing regime in the next 12 months, premium between carbonate and hydroxide will continue to stay volatile until the ex-China market successfully works to resolve the refining bottleneck.

Short battery metals basket. Aside from lithium, we recently downgraded our nickel price forecast on a growing class 1 surplus basis on the back of two key developments in the Indo-China regional markets. First, the fast-growing Indonesian matte output is leading to higher nickel sulphate output in China which is now creating a well-supplied class 1 market. Second, China's improved refined nickel supply on rising NPI and intermediate imports from Indonesia is restraining its appetite to import refined metal, in essence creating a softer ex-China market. On cobalt, the growing mixed hydroxide precipitate production – which holds both cobalt and nickel via the HPAL process - is contributing to an improved cobalt supply. Furthermore, the export ban resolution in DRC which is expected to now release ~15kt cobalt stockpile over the next 10-12 months is adding to an already oversupplied cobalt market, in our view. As a result, we lower our average price for cobalt to $32,000/t for 2023 and $30,000/t for 2024, and set the 12m target at $25,000/t. We see this environment of low pricing a good opportunity for the development of an ex-China supply chain which can provide raw materials compliant with the US's IRA and EU's Critical Raw Materials Act (CRMA). Finally, we close our long copper vs. short lithium trade to take profit of 13% and open a short battery metals basket (China lithium carbonate, nickel and cobalt metal) on expected oversupply led price correction.

No bottom in sight

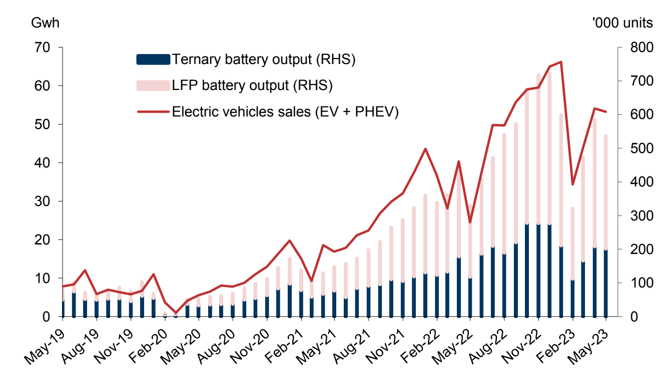
1. Structurally bearish fundamentals on oversupply. Battery metals prices have fallen significantly YTD, with China lithium carbonate prices declining 41%, nickel falling 29% and cobalt metal down 28% (see Exhibit 1). Whilst this correction in prices has been predominantly tied to a sluggish start to the year in China's EV sector (see Exhibit 2), we think that part of the sell-off also reflects respective markets' focus on strong supply trends across the complex. The already apparent fundamental shift from scarcity to a glut in the cobalt and nickel markets has led to continued price falls in search of tightening adjustments, whilst a sequential improvement in Chinese demand has led to a bounce in lithium prices where the oversupply is less pronounced for now. Going forward, on the demand side, we remain optimistic on EV sales in China for FY 2023 (8.5Mn units, +2Mn units y/y) on realised strategic EV price cuts, extension of tax exemptions and targeted stimulus support in rural areas (see Exhibit 3). However, we expect supply to sequentially soften the lithium market into surplus over H2 and continue to oversupply both the cobalt and nickel markets. Similar to the latest nickel update, we increase surplus for cobalt starting 2023, whereas for lithium 2023 we modestly lower our surplus before moving into a high surplus phase over 2024-25 (Exhibit 4). As a result, we expect battery metals prices to fall in H2, especially so for lithium where prices are still substantially above the top-end of cost curve which stands in contrast to an incremental necessity for supply rationing into 2024. We will, however, note that the long-term drivers of demand growth remain resilient as we raise our demand expectation from energy storage systems on renewable installation acceleration, particularly in China.

Exhibit 1: The YTD correction in battery prices has been driven by China's weak EV sector, we expect further fall on oversupply in H2



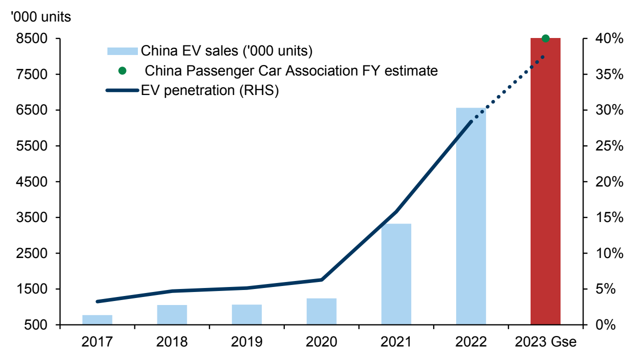
Source: Goldman Sachs Global Investment Research, Fastmarkets, SMM

Exhibit 2: China EV battery supply chain has recovered from Q1 lows but remains well below '22H2 trend



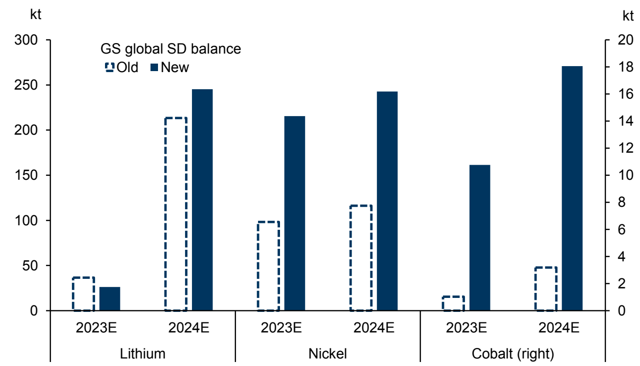
Source: Goldman Sachs Global Investment Research, CAAM, CPCA, Wind

Exhibit 3: We remain optimistic on EV sales in China for FY 2023 on realised strategic EV price cuts, extension of tax exemptions and targeted stimulus support in rural areas



Source: Goldman Sachs Global Investment Research, Wind

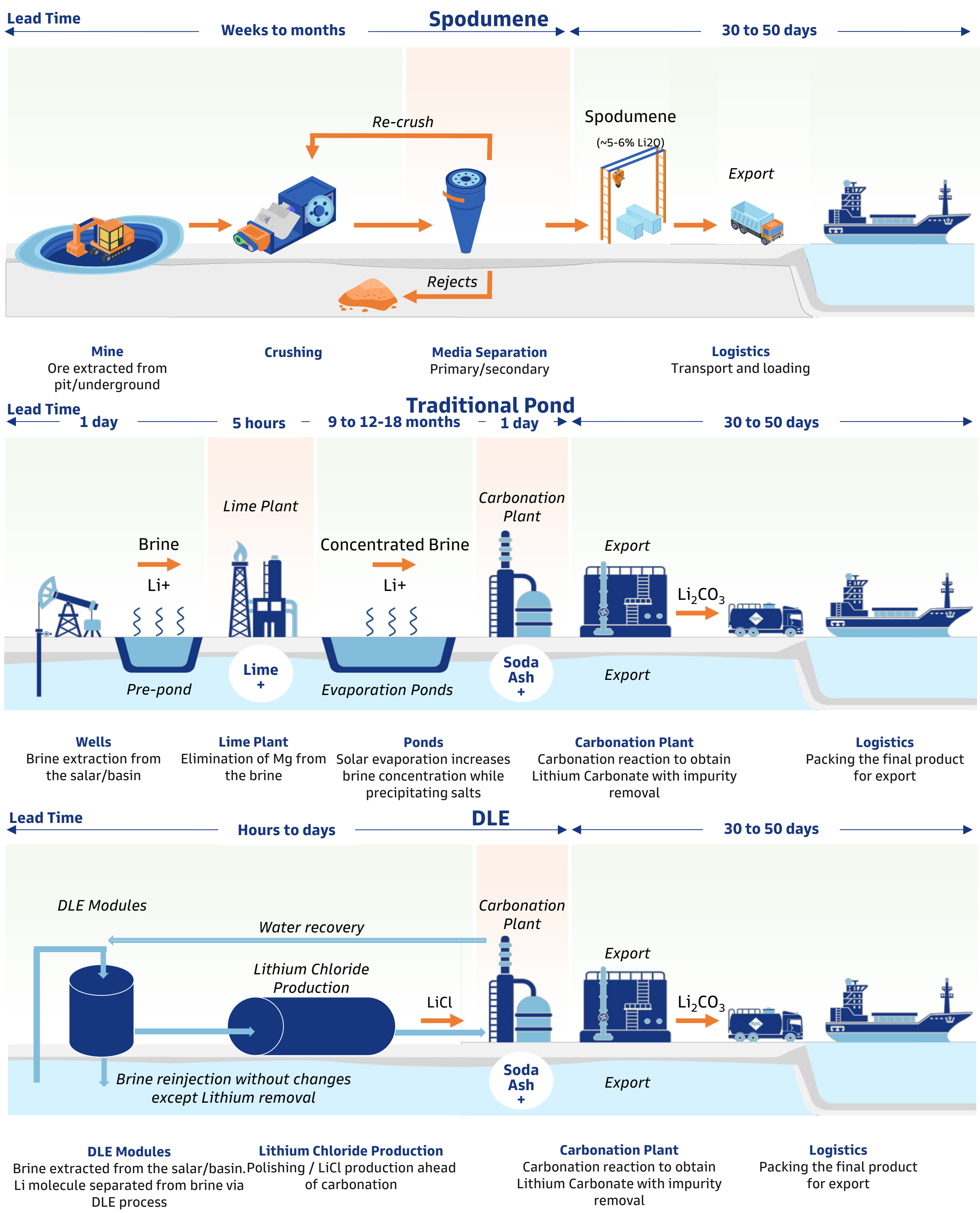
Exhibit 4: Strong supply growth is pushing battery metals into bigger surplus



Source: Goldman Sachs Global Investment Research

2. Supply growth remains resilient. We highlighted last year that the battery metals are at the beginning of a structural bear market as supply growth outpaces the strong demand. Over 2022, Chinese lithium chemical production grew +49%y/y driven by an increased raw material supply from both onshore and ex-China assets. Spodumene exports from Australia hit a 5-year high in 23Q1, and on our analysts' estimate spodumene projects continue to grow production with the aggregate ramp up in-line with their expectations. Low cost LatAm lithium brine projects have continued to ramp-up and whilst Chinese lepidolite projects have seen environmental issues, our China team expects limited risk to supply growth given developed water treatment technology which only marginally add to cost of production. We have also raised supply from Chinese owned (fully/partly) African assets (Arcadia, Bikita, Kamativi and DMCC) on successful project ramp-ups into the coming years, but note that infrastructure and transport challenges remain. Aside from lithium, the fast-growing Indonesian matte output is leading to higher nickel sulphate output in China which is now creating a well-supplied battery grade nickel market. On cobalt, the growing mixed hydroxide precipitate production – which holds both cobalt and nickel via the HPAL process - is contributing to an improved cobalt supply. This environment of low pricing in battery metals space is a good opportunity for the development of an ex-China supply chain which can provide raw materials compliant with the US's IRA and EU's Critical Raw Materials Act (CRMA), in our view.

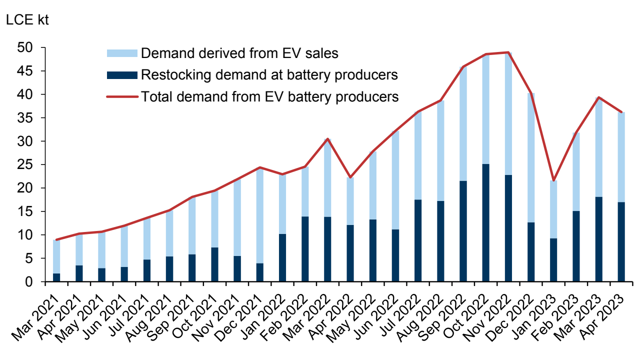
Exhibit 5: Lithium brine supply chain can see revolutionary change with DLE



Source: Goldman Sachs Global Investment Research, Company data

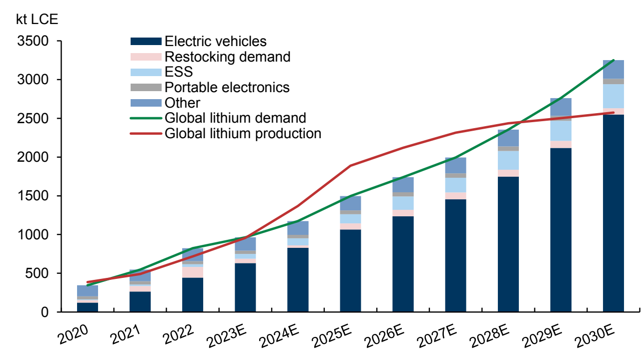
3. Lithium: Further medium-term downside. Over the past month, lithium prices have bounced back on signs of recovery in the Chinese EV market. Whist we expect recovering demand to support lithium prices in the near-term, we continue to see a supply led correction over '23H2. On our estimates, global spodumene is still the main driver of incremental raw material supply growth, on track to add 500kt LCE cumulative capacity over 2023-24, and China's lepidolite capacity will likely rise over 200kty LCE by end of next year, with total refined supply growing 321kty LCE on average. This compares against an average 226kty LCE of battery demand growth over the same period. In this context, we continue to see substantial downside to lithium prices, with our 12m target for China battery grade lithium carbonate (ex-VAT) at $15,000/t LCE but raise the 2024 average to $15,331/t LCE (vs. $11,000/t LCE prev.) on the higher cost for non-integrated projects and the lengthy quality check timeline for new feedstock by converters. However, if the boost driven by government measures and lower EV prices does not generate high EV sales in China over the next months, it will lead to an earlier sell-off in lithium prices exacerbated by high lithium inventories in EV supply chain, in our view. Equally, we note that the refining shift from carbonate to hydroxide will likely require an attractive conversion premium and set the 12m target for Asia CIF lithium hydroxide at $19,000/t. Whilst we think prices will move to a floor pricing regime in the next 12 months, premium between carbonate and hydroxide will continue to stay volatile until ex-China market successfully works to resolve the refining bottleneck.

Exhibit 6: Lithium demand in China is yet to return to pre-'22Q1 trend



Source: Goldman Sachs Global Investment Research, Wind, SMM

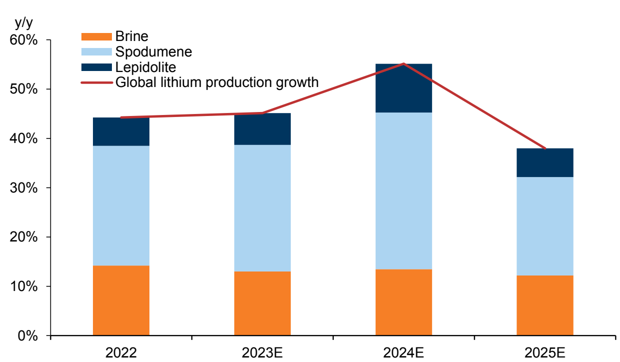
Exhibit 7: We continue to see massive supply response outpacing the significant demand growth in the coming years



Source: Goldman Sachs Global Investment Research, SMM, Woodmac, Company data, BNEF

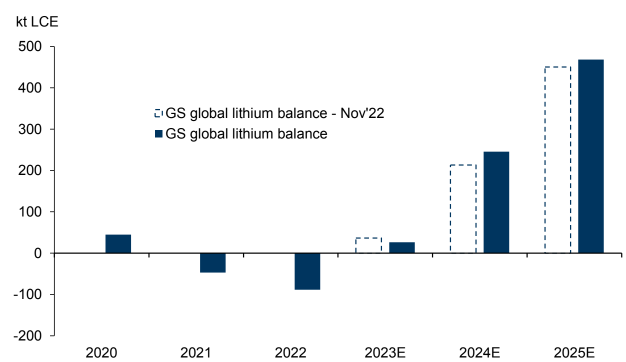
Exhibit 8: Globally spodumene is the main driver of raw material supply growth, on track to add 500kt LCE cumulative capacity over 2023-24

Global lithium production (pre-disruption adjustment)



Source: Goldman Sachs Global Investment Research, Company data, Woodmac, BNEF

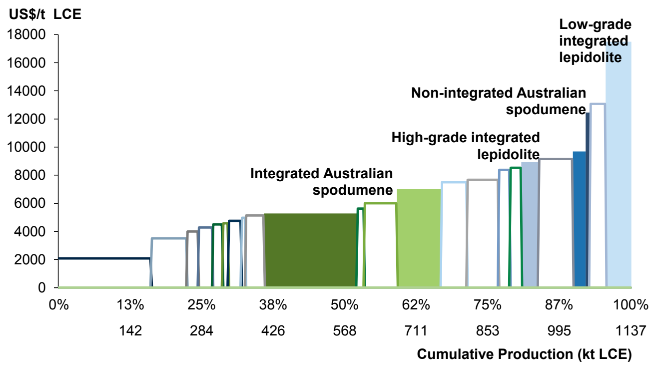
Exhibit 9: An increasingly oversupplied market is likely to lead to a regime of low prices over the coming years



Source: Goldman Sachs Global Investment Research, Woodmac, SMM, Data compiled by Goldman Sachs Global Investment Research

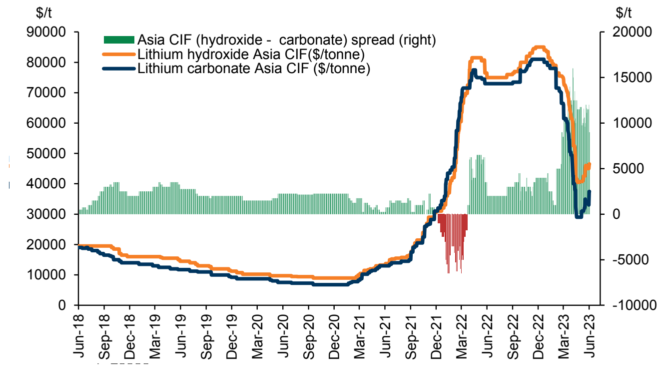
Exhibit 10: Prices are still substantially above the top-end of the cost curve in contrast to an incremental necessity for supply rationing into 2024

2024 cost curve, excluding royalties and VAT



Source: Goldman Sachs Global Investment Research, Data compiled by Goldman Sachs Global Investment Research, Woodmac

Exhibit 11: The hydroxide vs. carbonate premium will remain volatile but should converge to 2022 levels in the next 6-12 months



Source: Goldman Sachs Global Investment Research, Fastmarkets

Exhibit 12: GS Lithium supply-demand balance

Li ('000 tonnes LCE)	2020	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Global demand											
Consumption - batteries	**188**	**341**	**544**	**765**	**995**	**1268**	**1504**	**1750**	**2105**	**2505**	**3002**
% change y/y	*35%*	*82%*	*59%*	*41%*	*30%*	*27%*	*19%*	*16%*	*20%*	*19%*	*20%*
EV	103	231	395	569	744	963	1116	1310	1561	1884	2286
ESS	11	18	34	63	91	119	174	190	242	258	312
E-buses, two-wheeler EVs	20	33	48	64	84	102	120	146	187	235	263
Portable electronics	40	42	41	41	44	48	52	56	60	64	69
Other	14	17	25	28	32	36	42	48	55	64	73
Consumption - ex batteries	**128**	**135**	**141**	**143**	**147**	**150**	**154**	**157**	**161**	**164**	**168**
% change y/y	*-4%*	*5%*	*5%*	*2%*	*3%*	*2%*	*2%*	*2%*	*2%*	*2%*	*2%*
Ceramics	32	33	35	36	37	38	39	40	41	42	43
Glass-ceramics	26	27	29	30	31	31	32	33	34	35	36
Other	70	74	77	78	80	81	83	84	86	87	89
Restocking demand	**30**	**73**	**138**	**55**	**32**	**78**	**83**	**88**	**89**	**91**	**80**
% of global demand		15%	20%	6%	3%	6%	5%	5%	4%	3%	3%
Global demand (excl. restocking)	**315**	**476**	**684**	**908**	**1142**	**1419**	**1658**	**1907**	**2265**	**2670**	**3170**
Global demand (incl. restocking)	**345**	**549**	**823**	**963**	**1174**	**1497**	**1740**	**1995**	**2354**	**2760**	**3250**
% change y/y	*22%*	*59%*	*50%*	*17%*	*22%*	*28%*	*16%*	*15%*	*18%*	*17%*	*18%*
Global Refined Supply											
Brine	**161**	**228**	**301**	**396**	**538**	**740**	**804**	**938**	**1006**	**1037**	**1070**
China	36	60	79	115	152	230	231	256	267	277	288
Ex-China	126	168	222	280	386	510	573	682	739	759	782
Spodumene	**197**	**254**	**378**	**566**	**903**	**1232**	**1417**	**1486**	**1532**	**1561**	**1593**
China	10	27	54	70	105	154	166	198	215	231	248
Ex-China	187	227	324	496	798	1078	1251	1288	1317	1329	1345
Other (Lepidolite or Clay)	**28**	**25**	**54**	**101**	**206**	**302**	**333**	**365**	**396**	**417**	**438**
China	28	25	54	101	206	297	318	340	361	382	403
Ex-China	0	0	0	0	0	5	15	25	35	35	35
World output	**386**	**507**	**732**	**1062**	**1648**	**2274**	**2554**	**2788**	**2933**	**3014**	**3101**
% change y/y		*31%*	*44%*	*45%*	*55%*	*38%*	*12%*	*9%*	*5%*	*3%*	*3%*
Total output (adj. for disruption)	**386**	**492**	**717**	**956**	**1368**	**1887**	**2120**	**2314**	**2435**	**2502**	**2574**
% change y/y		*27%*	*46%*	*33%*	*43%*	*38%*	*12%*	*9%*	*5%*	*3%*	*3%*
Battery Scrap Supply	**4**	**9**	**17**	**34**	**52**	**78**	**96**	**101**	**128**	**213**	**339**
% change y/y		*2%*	*2%*	*4%*	*4%*	*4%*	*5%*	*4%*	*5%*	*9%*	*13%*
Global Balance	**45**	**-47**	**-89**	**26**	**245**	**468**	**475**	**420**	**209**	**-45**	**-337**
as % of global supply	12%	-10%	-12%	3%	18%	25%	22%	18%	9%	-2%	-13%

Source: Goldman Sachs Global Investment Research, BNEF, Woodmac, Company data

Exhibit 13: GS battery-grade lithium price forecasts

	Asia lithium hydroxide CIF CJK ($/t)	Asia lithium carbonate CIF CJK ($/t)	China lithium carbonate (RMB/t)	China Lithium carbonate (USD/t, with VAT)	China Lithium carbonate (USD/t, excluding VAT)	China lithium hydroxide (RMB/t)	China lithium hydroxide (USD/t, with VAT)	China lithium hydroxide (USD/t, excluding VAT)
2018	19453	17916	120850	18507	16378	134384	20450	18097
2019	14158	11660	70246	10215	9040	84461	12296	10882
2020	9734	7750	44382	6416	5678	52228	7544	6676
2021	16119	15139	112974	17534	15517	106738	16566	14660
2022	70435	69011	455466	69333	61357	450056	66936	59235
2023E	*54502*	*48500*	*316872*	*45331*	*40116*	*337639*	*48299*	*42743*
2024E	*22074*	*18324*	*118750*	*17324*	*15331*	*133750*	*19518*	*17273*

Source: Goldman Sachs Global Investment Research, SMM, Fastmarkets

Ex-China supply update (Hugo Nicolaci)

A summary of the state of the key Australian upstream producers/projects.

1. Talison's (Tianqi/Albemarle/IGO) **Greenbushes** (1.45-1.45Mt spodumene concentrate production guidance in FY23E, from 1.14Mt in FY22) continues to grow production with the ramp up of the tailings retreatment project, while chemical grade plants 3 & 4 potentially add an incremental 520ktpa of capacity each from mid-late decade.

2. Pilbara Minerals' **Pilgangoora** is growing production to 600-620kt in FY23E (5.4%), ramping up to 680ktpa from 4QCY23 with a 100ktpa expansion, and with a further expansion to >1Mtpa by FY26E following FID earlier this year (with additional volumes currently unallocated).

3. At **Wodgina** (Mineral Resources/Albemarle), trains 1 & 2 (250ktpa 6% spodumene nameplate) started in March & July 2022 respectively, with train 3 (additional 250ktpa) scheduled to start production in FY24E.

4. At **Mt. Marion** (Mineral Resources/Ganfeng) is targeting completion of the planned two stage upgrade to ~600ktpa 6% spodumene (900ktpa mixed grade) in FY24E.

5. Allkem's **Mt Cattlin** spodumene project will have rebounding volumes through FY23, impacted through 2H22 by lower ore grades/recoveries. Construction activities at **James Bay** are targeted to commence in 2023 with commissioning of the 320ktpa spodumene project in 2024/25. On brine assets, **Olaroz Stage 2** has completed and is moving through commissioning with production now expected from mid-2023, taking Olaroz to ~40ktpa of lithium carbonate by mid-decade (before enhanced recovery works), while a Stage 3 expansion (not in GSe) could take Olaroz to ~100ktpa LCE (possibly including Cauchari). At **Sal de Vida** the two-stage 45ktpa carbonate project is under construction with first production from the 15ktpa Stage 1 expected in FY24E (development of Stage 2 to commence in parallel with Stage 1 commissioning).

6. Core Lithium's **Finniss** mine expects first spodumene sales in 2Q23, ramping up to ~180ktpa from 2024, with ~80% of production over the first four years sold to offtake partners (including Ganfeng Lithium and Sichuan Yahua). The company is looking to further resource growth across its exploration portfolio.

7. Liontown's **Kathleen Valley** is in construction, with completion and commissioning is targeted for mid-2024 on an expanded project potentially delivering ~580ktpa (GSe) of spodumene concentrate, with a planned expansion to 700ktpa of spodumene concentrate within six years, and offtakes in place for at least the first 5 years with LG Energy, Tesla and Ford covering ~80% of production.

8. Covalent's (SQM/Wesfarmers) **Mt. Holland** project is progressing, targeting production from FY24 and

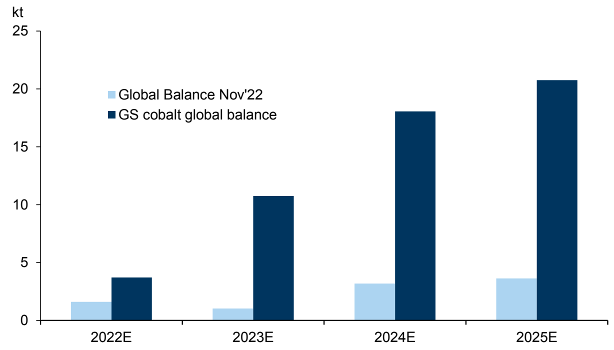
ramping up to ~380ktpa of spodumene concentrate, with studies to double capacity to ~760ktpa underway (not in GSe).

We find a number of additional projects and technology developers looking to utilise Direct Lithium Extraction (DLE) or support its implementation, with DLE increasingly being considered for the next wave of projects across both brownfield expansions at existing operations and emerging explorers/developers. We discussed the potential challenges around DLE implementation, where our view remains that with the ongoing efforts, DLE could be implemented between 2025-2030 in both Chile and Argentina.

Note: The above discussion of ex-China supply views are attributed Hugo Nicolaci who is an equity analyst covering Australia Basic Resources. All other views are those of the Commodities research team.

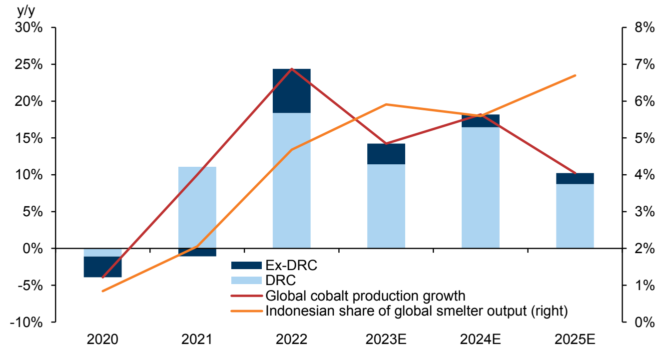
4. Cobalt: Structurally bearish on oversupply. Cobalt metal price has come down close to \$30,000/t on improved supply from DRC, growing output of Indonesian nickel-cobalt MHP alongside weak Chinese demand. We raise our surplus estimate over 2023-2025 (see Exhibit 14) on the back of two key developments on the supply front amid a EV weak demand environment. One, the resolution of royalties agreement between China Molybdenum Co (CMOC) with state-owned miner Gécamines at Tenke Fungurume DRC has lifted the export ban on Tenke produced cobalt. This resolution will lead to a release of ~15kt cobalt stockpile over the next 10-12 months, on our estimates. Two, the mixed hydroxide precipitate production – which holds both cobalt and nickel via the HPAL process - is contributing to an ex-DRC supply and should become popular amongst refiners due to the versatility of this feed. Combined, we see 25kty of supply growth in 2023-24 against an average demand growth of 20kty. In this context, we set a 12m target of \$25,000/t for cobalt metal, and further lower averages for 2023 to \$32,000/t (from \$48,375/t) and 2024/2025 to \$30,000 /t from (\$44,000/t). As we noted before, the rising share of no-cobalt LFP and high-nickel low cobalt batteries has also weighed on cobalt usage intensity - the average cobalt intensity in EVs has fallen to ~103g/kWh in 2022 (vs. 159g/kWh in 2020). Apart from batteries, whilst demand from the aerospace sector is strong, industrial sectors continue to suffer due to the cyclical headwinds. Over the long-term, we see an increased supply from secondary sources from US IRA and EU Critical Raw Materials Act (CRMA) as credit incentives for localisation and recycling.

Exhibit 14: Cobalt market remains structurally bearish on oversupply



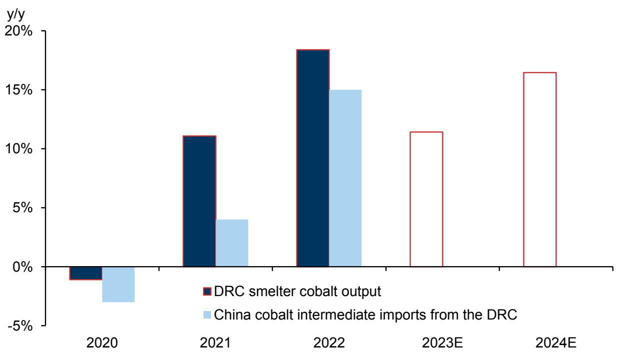
Source: Woodmac, BNEF, Fastmarkets, Goldman Sachs Global Investment Research

Exhibit 15: Indonesian mixed hydroxide precipitate production – which also contains cobalt the HPAL process - is contributing to an ex-DRC supply



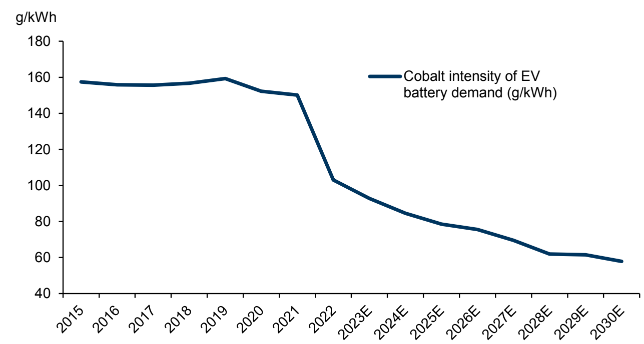
Source: Goldman Sachs Global Investment Research, Company data, SMM

Exhibit 16: High DRC production and low exports into China due to logistical disruptions has increased supply chain inventory



Source: SMM, ICSG, Goldman Sachs Global Investment Research

Exhibit 17: The rising share of no-cobalt LFP and high-nickel low cobalt batteries has weighed on EV cobalt usage intensity



Source: Goldman Sachs Global Investment Research

Exhibit 18: GS Cobalt supply-demand balance

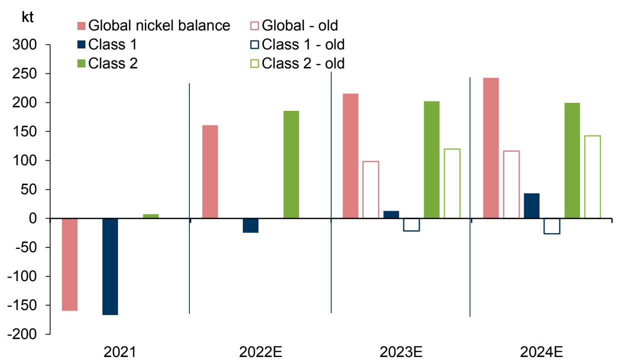
('000 tonnes)	2020	2021	2022E	2023E	2024E	2025E
Global demand						
Consumption - batteries	**82**	**110**	**117**	**136**	**153**	**173**
% of total	**61%**	**65%**	**66%**	**69%**	**70%**	**72%**
% change y/y	*9%*	*33%*	*7%*	*16%*	*13%*	*13%*
EV	20	43	52	68	81	97
ESS	2	3	3	5	6	8
E-buses, two-wheeler EVs	2	3	2	3	5	5
Portable electronics	56	57	56	56	57	59
Other	3	4	4	4	4	4
Consumption - ex batteries	**54**	**59**	**61**	**63**	**66**	**69**
% change y/y	*-13%*	*10%*	*3%*	*3%*	*5%*	*5%*
Superalloys	17	18	19	20	21	22
Tool materials	11	12	12	13	13	13
Other	26	29	29	30	31	34
Global consumption	**136**	**168**	**178**	**199**	**219**	**242**
% change y/y	-1%	24%	6%	12%	10%	11%
Global Production						
World smelter output	**140**	**154**	**192**	**219**	**259**	**285**
% change y/y	*-4%*	*10%*	*24%*	*14%*	*18%*	*10%*
DRC	96	111	139	161	197	220
% change y/y	*-2%*	*16%*	*25%*	*16%*	*22%*	*11%*
Ex- DRC	45	43	52	58	61	65
World refined output	**134**	**148**	**184**	**210**	**248**	**273**
World output (adj. for disruption)	**134**	**148**	**176**	**202**	**228**	**251**
% change y/y		*10%*	*19%*	*15%*	*13%*	*10%*
Battery Scrap Supply	**4**	**5**	**5**	**7**	**9**	**12**
% change y/y		*3%*	*3%*	*4%*	*4%*	*5%*
Global Balance	**2**	**-16**	**4**	**11**	**18**	**21**
as % of global supply	1%	-11%	2%	5%	8%	8%
Cash Prices (annual average)						
Current dollar ($/t)	33947	51119	69100	32000	30000	30000
Current dollar (c/lb)	1540	2319	3134	1452	1361	1361

Prices refer to Fastmarket cobalt metal standard grade

Source: Goldman Sachs Global Investment Research, Woodmac, BNEF

5. Nickel: End of class divide. We recently highlighted that the nickel market is set to move into a bigger surplus this year, on a gradual softening of the class 1 market (see Exhibit 19). Over the past 2 years, whilst the class 2 market moved into a surplus, the class 1 market remained tight as producers worked to ramp-up the class 2 to class 1 and MHP to sulphate conversion capacity in China. However, with the conversion of class 2 to class 1 nickel gathering pace, we expect the class 1 market to now move into a modest surplus. This crucial merger of nickel's class divide will be the key driver of surplus tendency in the class 1 market going forward, in our view. After two years of deficit this inflection in the class 1 to a surplus market points towards a continued downward trend in LME nickel prices; accordingly we recently lowered our average price for 2023 (GSe $20,525/t vs $22,750/t previously). This bearish path fundamental is anchored to two developments in the Indo-China regional markets. First, the fast growing Indonesian matte output (74kt YTD, +66% y/y) is leading to higher Chinese nickel sulphate output which is now creating a well-supplied class 1 market. Second, China's improved refined nickel supply on rising NPI and intermediate imports from Indonesia is restraining its appetite to import refined metal, in essence creating a softer ex-China market. Our latest 3/6/12M price targets ($19,000/17,000/16,000/t) reflect a move lower into the cost curve required to ration supply leading to an eventual balancing.

Exhibit 19: Nickel market to remain in a surplus into 2023 as supply continues to grow



Source: Goldman Sachs Global Investment Research, INSG

Exhibit 20: NPI-to-matte conversion is becoming the crucial Indonesian intermediate for battery grade nickel supply

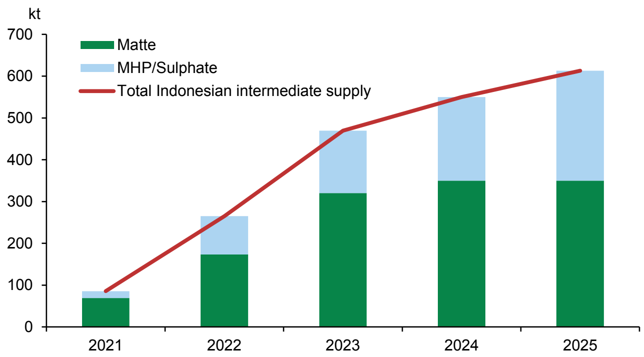
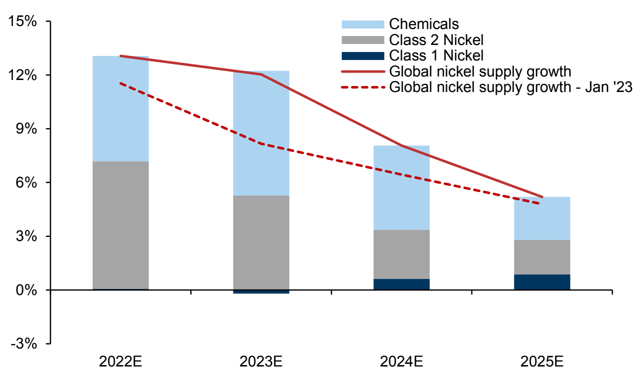
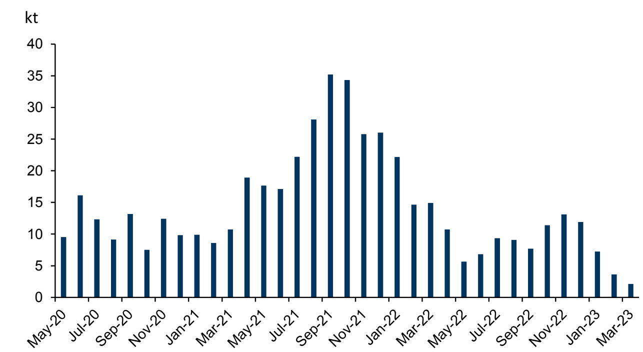


Exhibit 21: We expect a higher supply of nickel refined products due to an increase in chemicals output



Source: Woodmac, Goldman Sachs Global Investment Research

Exhibit 22: A well supplied domestic market on increasing Indo intermediate supply is weighing on refined import demand



Source: Goldman Sachs Global Investment Research, SMM

Exhibit 23: GS Global Nickel SD Model

('000 tonnes)	2016	2017	2018	2019	2020	2021	2022E	2023E	2024E	2025E
Consumption - DM										
US	136	156	164	154	134	139	137	151	188	249
% change y/y	2%	14%	5%	-6%	-13%	4%	-2%	11%	24%	32%
Europe	338	334	333	305	287	332	318	340	367	394
% change y/y	6%	-1%	0%	-8%	-9%	15%	-4%	7%	8%	7%
Japan	146	156	170	154	135	158	154	163	168	177
% change y/y	5%	7%	8%	-9%	-12%	17%	-3%	6%	4%	5%
Other DM	9	9	8	8	7	8	9	9	9	9
% change y/y	-7%	3%	-8%	-1%	-11%	7%	8%	2%	2%	2%
Sub- DM	630	655	675	621	563	637	617	662	731	828
% change y/y	5%	4%	3%	-8%	-9%	13%	-3%	7%	10%	13%
Consumption - EM										
China	1178	1199	1180	1345	1448	1507	1598	1795	1923	1999
Indonesia	3	55	170	164	206	383	395	443	485	528
China & Indonesia	1181	1254	1349	1510	1654	1890	1993	2237	2408	2527
% change y/y	15%	6%	8%	12%	10%	14%	5%	12%	8%	5%
Other EM	326	332	337	325	259	337	289	322	342	366
% change y/y	11%	2%	1%	-3%	-18%	30%	-14%	11%	6%	7%
Sub- EM	1506	1586	1686	1835	1913	2227	2282	2559	2750	2893
% change y/y	14%	5%	6%	9%	5%	16%	2%	12%	7%	5%
Global Consumption	**2136**	**2241**	**2361**	**2456**	**2477**	**2864**	**2899**	**3221**	**3481**	**3721**
% change y/y	11%	5%	5%	4%	1%	16%	1%	11%	8%	7%
World ex-China & Indonesia	955	987	1012	947	823	974	906	984	1073	1194
% change y/y	7%	3%	3%	-6%	-12%	18%	-7%	9%	9%	11%
Global Production										
Global Primary Production	**2048**	**2042**	**2186**	**2372**	**2530**	**2699**	**3051**	**3418**	**3694**	**3886**
% change y/y	*2%*	*0%*	*7%*	*9%*	*7%*	*7%*	*13%*	*12%*	*8%*	*5%*
China	**615**	**600**	685	810	771	747	869	1071	1186	1221
% change y/y	*-2%*	*10%*	*9%*	*18%*	*-4%*	*-3%*	*16%*	*23%*	*11%*	*3%*
Indonesia	**119**	**204**	290	386	639	907	1155	1296	1392	1508
% change y/y	*152%*	*70%*	*43%*	*33%*	*65%*	*42%*	*27%*	*12%*	*7%*	*8%*
ex-China & Indonesia	**1313**	**1237**	1211	1176	1120	1044	1028	1051	1116	1157
% change y/y	*-2%*	*-5%*	*-2%*	*-3%*	*-4%*	*-7%*	*-2%*	*2%*	*6%*	*4%*
Global Battery Scrap Supply					3	6	9	19	30	49
Global Balance	**-88**	**-200**	**-176**	**-84**	**56**	**-159**	**161**	**215**	**243**	**214**
Cash Prices (annual average)										
Current dollars ($/t)	9631	10414	13118	13944	13803	18474	28316	20525	18000	18000
Current dollars (c/lb)	437	472	595	632	626	838	1284	931	816	816

Source: Goldman Sachs Global Investment Research, Woodmac, SMM, BNEF

Appendix

Exhibit 24: GS Cobalt Green Demand Model

GS Cobalt Green Demand Model

000 tonnes	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Europe	**9**	**8**	**9**	**10**	**13**	**18**	**22**	**27**	**34**	**39**
EV	9	8	9	9	12	16	21	24	31	36
ESS	1	0	1	1	1	2	2	3	3	3
YoY growth	*122%*	*-14%*	*14%*	*11%*	*31%*	*35%*	*25%*	*19%*	*27%*	*17%*
US	**4**	**4**	**7**	**12**	**21**	**29**	**35**	**37**	**43**	**50**
EV	4	4	6	11	19	26	32	34	39	45
ESS	0	0	0	1	2	2	3	3	4	4
YoY growth	*41%*	*-1%*	*51%*	*74%*	*77%*	*38%*	*21%*	*7%*	*15%*	*16%*
China	**27**	**39**	**52**	**60**	**65**	**68**	**69**	**73**	**83**	**92**
EV	26	37	49	55	60	62	64	66	76	84
ESS	2	2	4	4	5	6	6	7	7	7
YoY growth	*127%*	*42%*	*34%*	*14%*	*9%*	*5%*	*2%*	*5%*	*15%*	*10%*
RoW	**4**	**4**	**5**	**6**	**6**	**6**	**7**	**10**	**14**	**17**
EV	4	4	4	5	6	6	7	9	13	16
ESS	0	0	0	0	0	1	1	1	1	1
YoY growth	*95%*	*-14%*	*20%*	*26%*	*5%*	*1%*	*22%*	*36%*	*41%*	*21%*
Global cobalt green demand	**49**	**57**	**76**	**92**	**110**	**127**	**141**	**154**	**182**	**208**
EV	43	52	68	81	97	110	123	133	159	182
ESS	3	3	5	6	8	10	11	14	15	16
E-buses, two-wheeler EVs	3	2	3	5	5	6	7	8	8	10
YoY growth	*105%*	*18%*	*32%*	*21%*	*20%*	*15%*	*11%*	*10%*	*18%*	*14%*

Source: Goldman Sachs Global Investment Research

Exhibit 25: GS Lithium Green Demand Model

GS Lithium Green Demand Model

'000 tonnes LCE	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Europe	**51**	**63**	**80**	**97**	**137**	**192**	**252**	**328**	**416**	**518**
EV	47	58	72	87	122	166	220	284	366	456
ESS	4	5	8	11	15	26	32	44	50	62
YoY growth	*132%*	*23%*	*27%*	*22%*	*41%*	*40%*	*31%*	*30%*	*27%*	*25%*
US	**24**	**35**	**58**	**112**	**213**	**305**	**389**	**457**	**525**	**650**
EV	23	32	53	100	190	264	339	396	462	572
ESS	2	3	6	12	23	41	49	61	63	78
YoY growth	*47%*	*41%*	*69%*	*92%*	*90%*	*43%*	*27%*	*18%*	*15%*	*24%*
China	**150**	**302**	**455**	**571**	**669**	**727**	**776**	**893**	**1027**	**1206**
EV	139	279	409	508	596	629	678	773	903	1061
ESS	11	24	45	62	74	98	98	120	124	145
YoY growth	*137%*	*102%*	*50%*	*26%*	*17%*	*9%*	*7%*	*15%*	*15%*	*17%*
RoW	**24**	**29**	**39**	**55**	**62**	**65**	**83**	**124**	**174**	**224**
EV	22	27	35	49	55	56	73	107	153	198
ESS	2	2	4	6	7	9	11	17	21	27
YoY growth	*104%*	*23%*	*34%*	*39%*	*13%*	*5%*	*28%*	*49%*	*41%*	*29%*
Global lithium green demand	**282**	**477**	**696**	**919**	**1184**	**1410**	**1646**	**1989**	**2377**	**2861**
EV	231	395	569	744	963	1116	1310	1561	1884	2286
ESS	18	34	63	91	119	174	190	242	258	312
E-buses, two-wheeler EVs	33	48	64	84	102	120	146	187	235	263
YoY growth	*112%*	*69%*	*46%*	*32%*	*29%*	*19%*	*17%*	*21%*	*19%*	*20%*

Source: Goldman Sachs Global Investment Research

Exhibit 26: GS Nickel Green Demand Model

GS Nickel Green Demand Model

000 tonnes	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Europe	**45**	**45**	**54**	**65**	**91**	**129**	**170**	**215**	**268**	**318**
EV	37	36	45	56	79	113	150	192	243	292
ESS	2	2	2	2	3	5	6	8	8	9
Wind	6	7	6	7	9	11	14	16	16	17
YoY growth	*93%*	*-1%*	*21%*	*22%*	*38%*	*42%*	*32%*	*27%*	*24%*	*19%*
US	**24**	**24**	**37**	**68**	**127**	**185**	**236**	**275**	**315**	**375**
EV	18	20	33	63	118	174	224	261	302	360
ESS	1	1	2	3	5	8	9	10	11	12
Wind	6	4	3	3	4	3	3	3	3	3
YoY growth	*36%*	*-1%*	*55%*	*82%*	*86%*	*46%*	*27%*	*16%*	*15%*	*19%*
China	**136**	**202**	**301**	**372**	**433**	**481**	**524**	**594**	**679**	**766**
EV	114	179	262	326	381	423	464	527	608	691
ESS	5	8	12	13	15	18	17	20	21	22
Wind	17	15	27	32	38	39	43	47	50	53
YoY growth	*61%*	*48%*	*49%*	*24%*	*16%*	*11%*	*9%*	*13%*	*14%*	*13%*
RoW	**55**	**50**	**67**	**82**	**94**	**101**	**119**	**148**	**182**	**212**
EV	17	17	22	30	34	37	49	71	101	125
ESS	1	1	1	1	1	2	2	3	4	4
Wind	37	32	44	50	58	62	68	74	78	83
YoY growth	*0%*	*-10%*	*36%*	*21%*	*15%*	*7%*	*18%*	*24%*	*23%*	*16%*
Global green demand nickel	**260**	**320**	**460**	**587**	**744**	**896**	**1049**	**1232**	**1444**	**1671**
EV	168	223	323	429	554	686	813	969	1167	1369
ESS	8	11	16	20	24	32	33	41	44	48
E-buses, two-wheeler EVs	18	29	40	47	58	62	74	82	86	99
Wind	65	58	80	92	109	116	128	139	147	155
YoY growth	*44%*	*23%*	*43%*	*28%*	*27%*	*20%*	*17%*	*17%*	*17%*	*16%*

Source: Goldman Sachs Global Investment Research

Related Research

Green Metals: Demand revolution gathers momentum

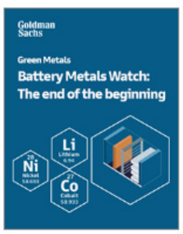
16 Mar 2023

While many investors can see the structural demand surge for green metals, we believe its true impact on fundamentals remains underappreciated in the market today. Green demand is at the heart of the metal supercycle—and the drivers are diversifying. The most important development over the last year has been the rise in strategic industrial competition between China, Europe, and the US as they race to attract clean tech capital amid the rise of programs like the US IRA and the European Green Deal Industrial Plan. Our commodities team continues to see China driving the majority of green demand near-term, but see rising demand alongside policy support in other regions. For copper alone—the "oil" of the green economy—our strategists see US and European share growing to ~43% of the 6.8 Mt global green demand they estimate by the end of the decade, from ~30% share today.

Battery Metals Watch: Lithium's delayed decline

9 Nov 2022

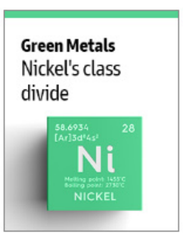
Stimulus-led restocking implies lithium prices will remain high near term. In Q2 this year, we called for peaking battery metals prices as mounting supply responses were set to catch up to the accelerating demand. Since then, the cobalt price has fallen 35% and nickel is down 40%, but onshore lithium prices have risen (24%) to record levels. For lithium, whilst the supply response we had estimated has continued to gather pace, as evidenced by surging Chinese lithium carbonate output, LFP battery expansion related restocking demand and higher-than-expected EV sales in China due to the exceptional stimulus have kept the market tighter than expected.



Battery Metals Watch: The end of the beginning

29 May 2022

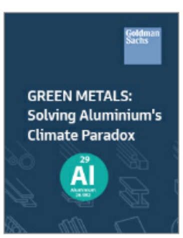
Battery metals – cobalt, lithium and nickel – will power the green industrial revolution, facing a wave of demand comparable to that of copper and iron ore during China's rapid growth in the 2000's. With climate change top of mind, investors are fully aware that battery metals will play a crucial role in the 21st century global economy, just as bulk and base metals did before them. Yet despite this exponential demand profile, we see the battery metals bull market as over for now.



Nickel's class divide

28 Apr 2022

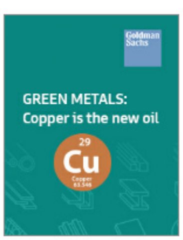
At the beginning of the year, nickel's place within green metals was as a key competitor in the race for mineral dominance of energy storage. Now, it sits at the intersection of Europe's push for decarbonisation and energy independence. At the heart of Europe's strategy lies its desire to rapidly electrify its transportation sector - a source of 20% of its emissions and c.2mb/d of Russian oil imports.



Solving Aluminium's Climate Paradox

20 Jun 2021

At the heart of the coming surge in green aluminium demand lies a paradox: aluminium is a key input required to produce decarbonising technologies like EV's and solar power, yet its own production is very carbon intensive, generating 2% of all global emissions. This paradox begs the question: how can we secure enough aluminium to effectively decarbonise, while keeping the climate impact of the path to net zero to a minimum? In our view, the resolution of this paradox will drive a structural bull market in aluminium over the next half decade, driven by the necessity to grow supply to meet green demand while cutting emissions to prevent a climate catastrophe.



Copper is the new oil

13 Apr 2021

The critical role copper will play in achieving the Paris climate goals cannot be overstated. As the most cost-effective conductive material, copper sits at the heart of capturing, storing and transporting these new sources of energy. Leveraging equity analysis across EVs, wind, solar, and battery technology, we estimate that by 2030, copper demand will grow nearly 600%. Yet the copper market as it currently stands is not prepared for this demand, and producers are leery of committing to new supply. We significantly increase our price forecasts and explore the implications across the metals and mining complex.

Disclosure Appendix

Reg AC

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